RNS INSTITUTE OF TECHNOLOGY

Department of Computer Science & Engineering

Major Project (BCS685)Phase-2:Review-1

ReflectAI: A Depth-Adaptive Socratic Reflection Engine for Longitudinal AI Journaling

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Designation

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AGENDA

1. Introduction
2. Literature Survey
3. Problem Statement
4. Objectives
5. Methodology/System Architecture
6. Results (showcase atleast 60-70% work)
7. Timeline/Plan
8. Future Scope
9. Patent/ Publication Details
10. References

INTRODUCTION

- **Motivation.** Reflective, free-form journaling is a well-evidenced tool for emotional processing and self-awareness, but adherence collapses fast without prompts, structure, or feedback — most people stop within days.
- **What ReflectAI is.** A full-stack, AI-powered smart-journaling and digital-health platform (in-app branded "Equoria") that pairs free-form journaling with a Socratic-style AI reflection partner, automatic mood analytics, and passive physical-health correlation.
- **How it works.** Every entry is mood-tagged and timestamped; an LLM-driven reflection engine asks context-aware follow-up questions grounded in the user's own journal history, with a safety-checked heuristic fallback so the experience degrades gracefully instead of breaking.
- **Health context.** A native iOS companion app streams Apple HealthKit signals (sleep, steps, heart rate, stress) to the platform, which the Retrospect module correlates against mood over time.
- **Engineering scope.** Built on React + Vite, Node.js/Express + MongoDB, and SwiftUI/HealthKit, with production-grade security — JWT rotation, 2FA, encryption at rest, rate limiting — engineered in from the start, not bolted on later.
- **This review.** Phase-2 demonstrates a working end-to-end system: authentication, journaling, AI-guided reflection, mood/retrospective analytics, Apple Health sync, and a fully customizable interface.

LITERATURE SURVEY

- **LLM-based journaling & chatbot efficacy [1]–[7].** Multiple 2023–2026 systematic reviews and meta-analyses confirm AI conversational agents produce measurable reductions in depressive/anxiety symptoms (effect sizes $g \approx 0.27$–0.82), and that LLM-driven journaling is an emerging, distinct UX pattern from static chat.

- **Socratic / CBT-style dialogue [10]–[12].** 2024–2025 studies show LLMs can reliably execute Socratic questioning and cognitive-restructuring protocols used in CBT, validated by mental-health professionals reviewing real transcripts — directly informs ReflectAI's follow-up-question engine.

- **Digital phenotyping & mood prediction [8]–[9].** Passive smartphone/wearable signals (sleep, steps, mobility) reliably classify and predict mood and depression risk, motivating ReflectAI's Health–Retrospect correlation.

- **Foundations of journaling itself [13].** Pennebaker-tradition expressive-writing research remains the strongest evidence base for why guided journaling improves psychological and physical health outcomes.

- **NLP mood/emotion classification [14].** Transformer- and LSTM-based classifiers now reach ≈90% accuracy on text mood classification, informing ReflectAI's mood-tag and theme-extraction pipeline.

- **Privacy & security posture [15].** 2024 CHI research found mental-health apps are frequently under-secured (1,575 vulnerabilities across just 10 audited apps) — directly motivated ReflectAI's own multi-round security-hardening pass.

- *Conclusion: the literature strongly supports each individual piece — journaling helps, LLMs can deliver CBT/Socratic-style support, and passive health signals predict mood — but very few systems combine all three with production-grade security. ReflectAI is designed to close exactly that gap.*

PROBLEM STATEMENT

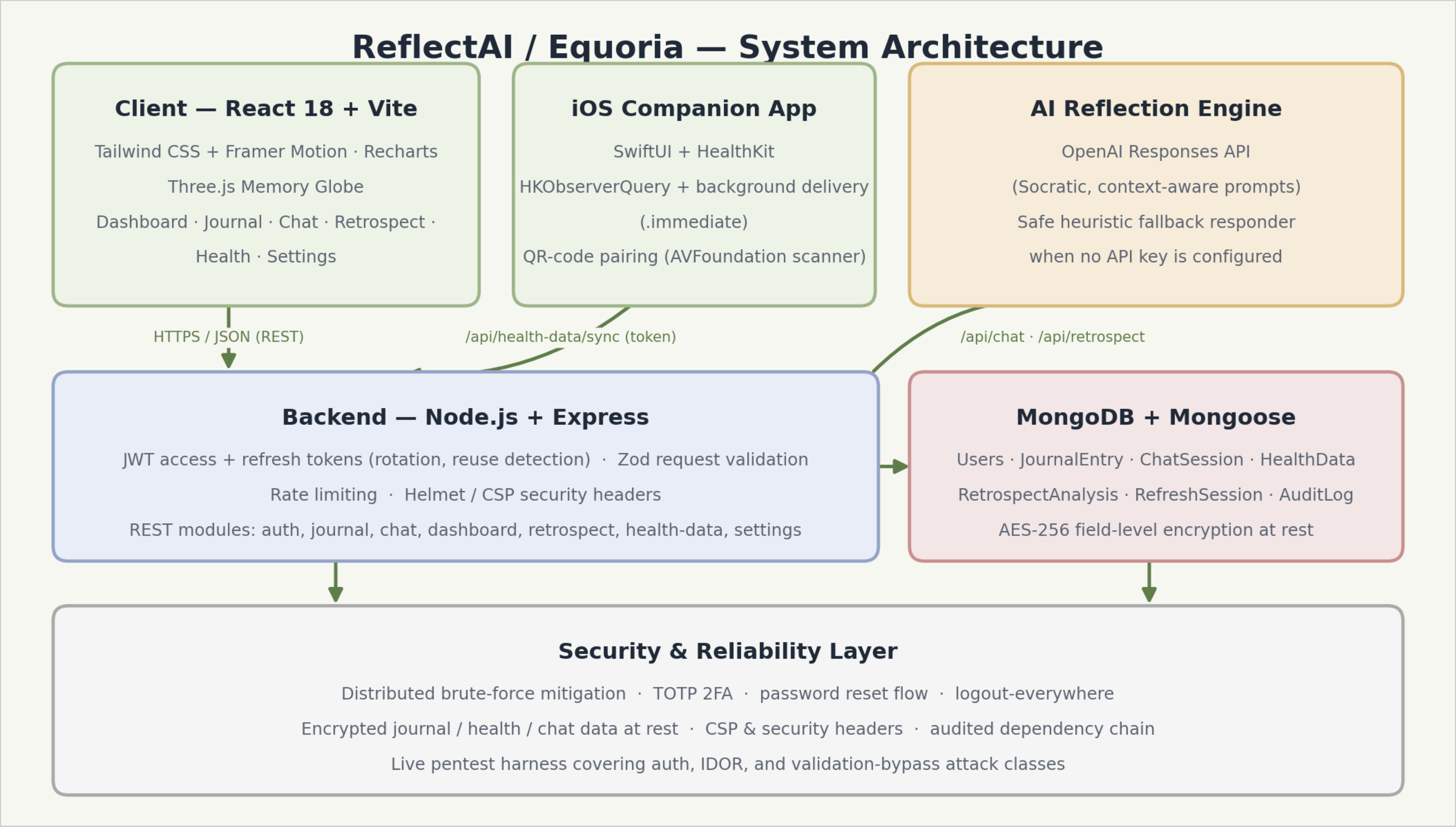
- Most journaling apps are passive notebooks with no feedback loop, while most AI mental-health chatbots are stateless, one-off conversations disconnected from a user's own writing history and physical well-being. Users are left to manually notice their own emotional patterns, with nothing connecting mood swings to sleep, activity, or workload, and no safe, always-available prompt to go deeper than a surface-level entry — resulting in shallow, inconsistent journaling and missed early signals of stress or burnout.

- ***There is no accessible, secure, single platform that unifies guided reflective journaling, AI-driven Socratic follow-up, and objective physical-health context into one continuously learning feedback loop.***

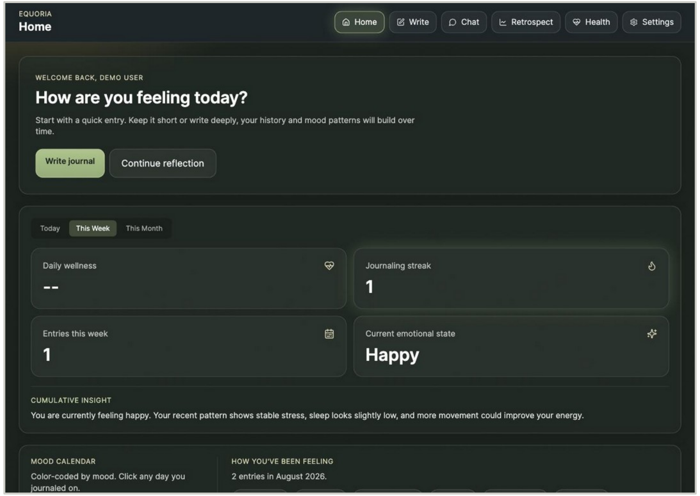
OBJECTIVES

- Build a secure, full-stack smart-journaling platform where every entry carries a mood tag and, optionally, linked physical-health context.
- Design an AI reflection engine that asks context-aware, Socratic-style follow-up questions grounded in the user's own journal history, with a safe non-AI fallback so the experience never fully breaks.
- Surface emotional patterns automatically — recurring themes, behavioral loops, mood-vs-health correlations — instead of requiring the user to notice them manually.
- Integrate real physical-health data (steps, sleep, heart rate, stress) via a native iOS HealthKit companion app with near-real-time background sync.
- Engineer the platform to a production security bar: JWT rotation, 2FA, rate limiting, encryption at rest, and a live, self-authored penetration-test harness.
- Give users full control over the interface's look and feel through simple accent-color sliders, with zero code changes required.

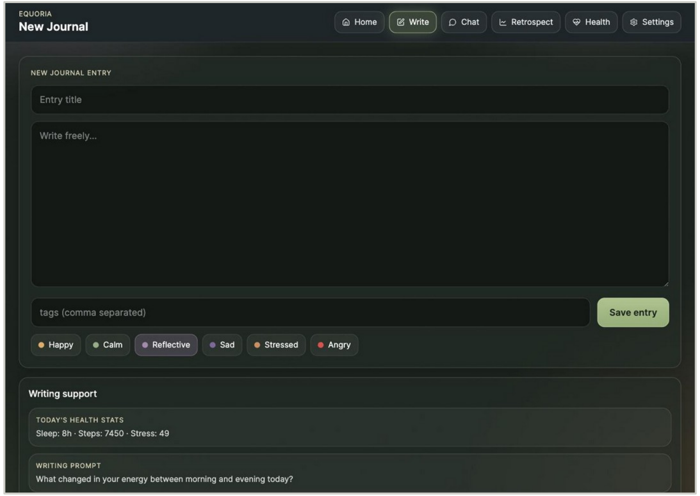
METHODOLOGY/SYSTEM ARCHITECTURE



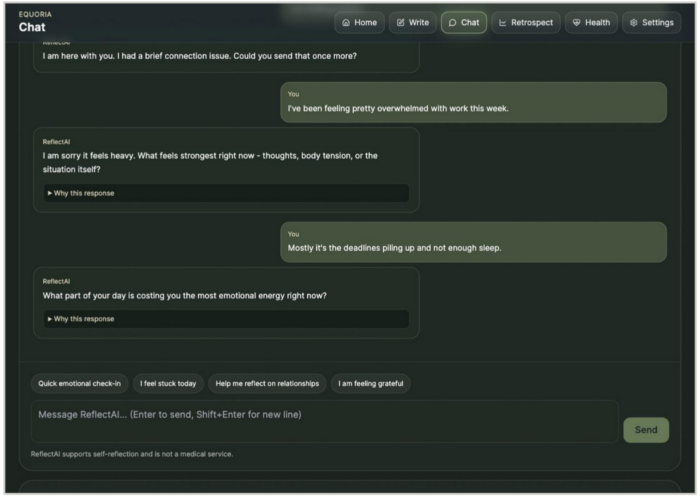
RESULTS I — JOURNALING & AI REFLECTION



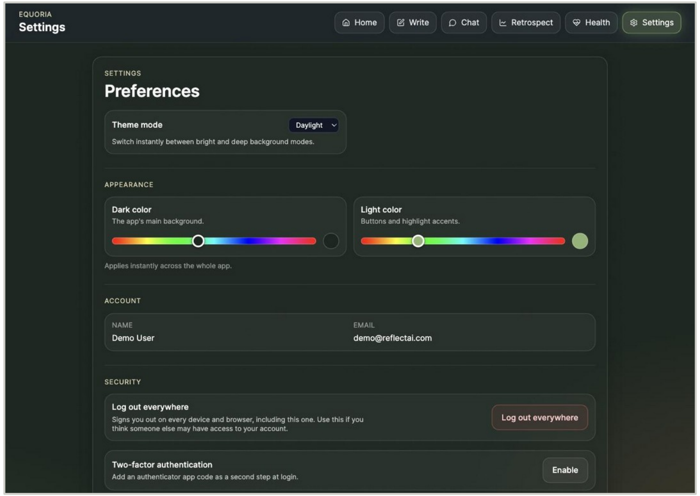
Dashboard — mood calendar, streaks, cumulative insight



New Journal Entry — mood tagging + writing support



AI Reflection Chat — Socratic follow-up questioning

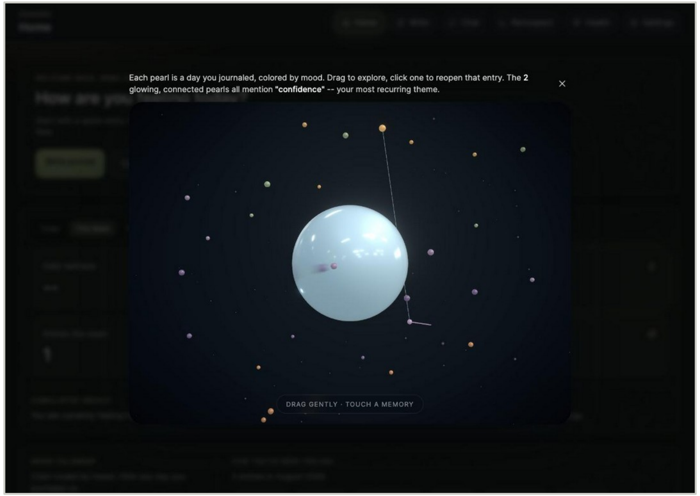


Settings — live dark/light accent-color sliders

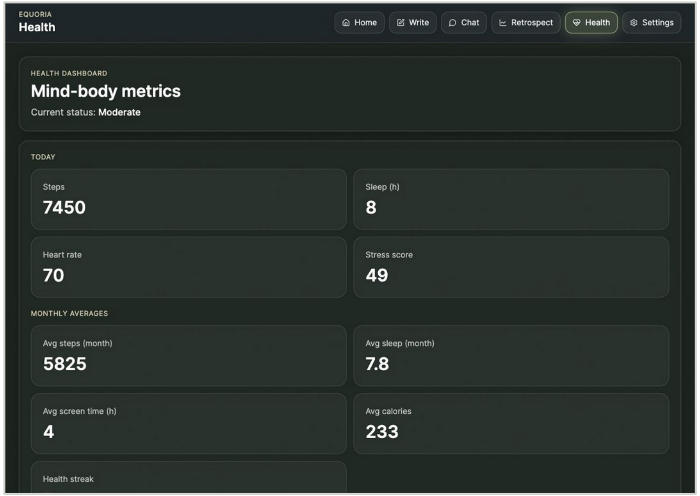
RESULTS II — ANALYTICS & HEALTH SYNC

Retrospect — emotional-timeline chart + recurring themes

Apple Health sync — weekly steps / stress / sleep trends



Memory Globe — 3D constellation of journaled days



Health dashboard — today's metrics + monthly averages

TIMELINE/PLAN

Phase	Duration	Key Deliverables	Status
1 — Planning & Design	Weeks 1–3	Requirement analysis, literature survey, system architecture, DB schema design	**Complete**
2 — Core Platform	Weeks 4–7	JWT/2FA auth, journal CRUD, mood tagging, dashboard, MongoDB models	**Complete**
3 — AI Reflection Engine	Weeks 8–9	OpenAI-integrated Socratic chat, heuristic fallback, anti-repetition logic	**Complete**
4 — Analytics & Health Sync	Weeks 10–11	Retrospect analytics, theme extraction, iOS HealthKit app, QR pairing	**Complete**
5 — Security Hardening	Week 12	Rate limiting, encryption at rest, pentest harness, 6 major risk fixes	**Complete**
6 — UI/UX Polish (current)	Week 13 (this review)	Memory Globe, custom theming, layout overhaul, live QA pass	**In progress (~65%)**
7 — Final Testing & Deployment	Weeks 14–16	End-to-end test suite, deployment, extended analytics, demo prep	**Planned**

FUTURE SCOPE

- Nightingale Rose Chart for richer multi-dimensional mood/emotion visualization (season-over-season emotional composition) — currently in design discussion.
- Android companion app to reach parity with the existing iOS HealthKit integration, via the Health Connect API.
- Deeper sentiment/emotion NLP — transformer-based multi-label emotion classification — replacing the current keyword-based theme extraction.
- Optional clinician/therapist-facing dashboard for supervised use cases, gated behind explicit user sharing consent.
- On-device / federated personalization to reduce reliance on a hosted LLM and strengthen privacy guarantees further.
- Multi-language journaling and AI reflection support.
- Adaptive, data-driven habit nudges — building on the existing streak system — informed by the digital-phenotyping literature [8], [9].

PATENT/PUBLICATION DETAILS

- No patents have been filed and no publications have been made for this project at this stage of development (Phase-2 Review).

- A conference/journal manuscript summarizing the Socratic-reflection engine and the mood–health correlation methodology is planned for submission after Phase-3 completion and full evaluation.

- **Novelty being explored for potential filing:** the heuristic-fallback anti-repetition mechanism that keeps AI-guided reflection safe, contextual, and non-repetitive without requiring a live LLM connection.

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